CHITTAGONG UNIVERSITY OF ENGINEERING & TECHNOLOGY



DEPARTMENT OF BIOMEDICAL ENGINEERING

PROJECT REPORT

COURSE TITLE: TELEMEDICINE & HEALTHCARE SESSIONAL

COURSE CODE: BME 416

IOT Enable Advanced Environmental Monitoring System for Hospitals

SUBMITTED TO:

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Problem statement:

Hospitals are environments where health, safety, and comfort must be maintained at the highest standards, particularly in critical care areas such as operating theaters, intensive care units (ICUs), and isolation rooms. These spaces demand strict environmental control to minimize the risk of infection, enhance patient recovery, and ensure the efficiency of medical staff. Even slight deviations in environmental conditions can have significant consequences, as patients in these areas are often highly vulnerable, and medical procedures are sensitive to external factors.

Among the most important parameters influencing hospital environments are temperature, humidity, and overall air quality. Maintaining an optimal temperature ensures patient comfort, reduces stress on the body, and prevents microbial growth. Similarly, humidity control plays a vital role in reducing the survival of pathogens, preventing equipment malfunction, and maintaining air sterility. Moreover, poor air quality, often due to particulate matter or toxic gases, can increase the risk of infections and compromise sterile conditions in surgical and intensive care units.

To address these challenges, this project aims to design and implement a comprehensive environmental monitoring system that integrates multiple sensors to measure temperature, humidity, and air quality in real time. The system will continuously monitor these critical parameters and provide instant alerts when deviations from recommended health and safety standards occur. Such a system not only enhances patient safety but also supports healthcare personnel by ensuring that they operate in a controlled and safe environment. Ultimately, this technology can contribute to improved patient outcomes, reduced hospital-acquired infections, and more efficient healthcare delivery.

Objectives:

- To implement a sensor network capable of monitoring key environmental parameters in hospital settings.
- To ensure that hospitals maintain appropriate conditions, especially in high-risk areas like operating rooms and isolation units.

- To create an automated LED alert system that will notify healthcare personnel of critical deviations in environmental parameters.

Instrument lists:

- ESP32 Development Board
- DHT22 temperature & humidity sensor
- MP 135 air quality sensor (smoke, alcohol, carbon monoxide)
- Breadboard
- Wire
- Power source
- Blynk software

Block Diagram:

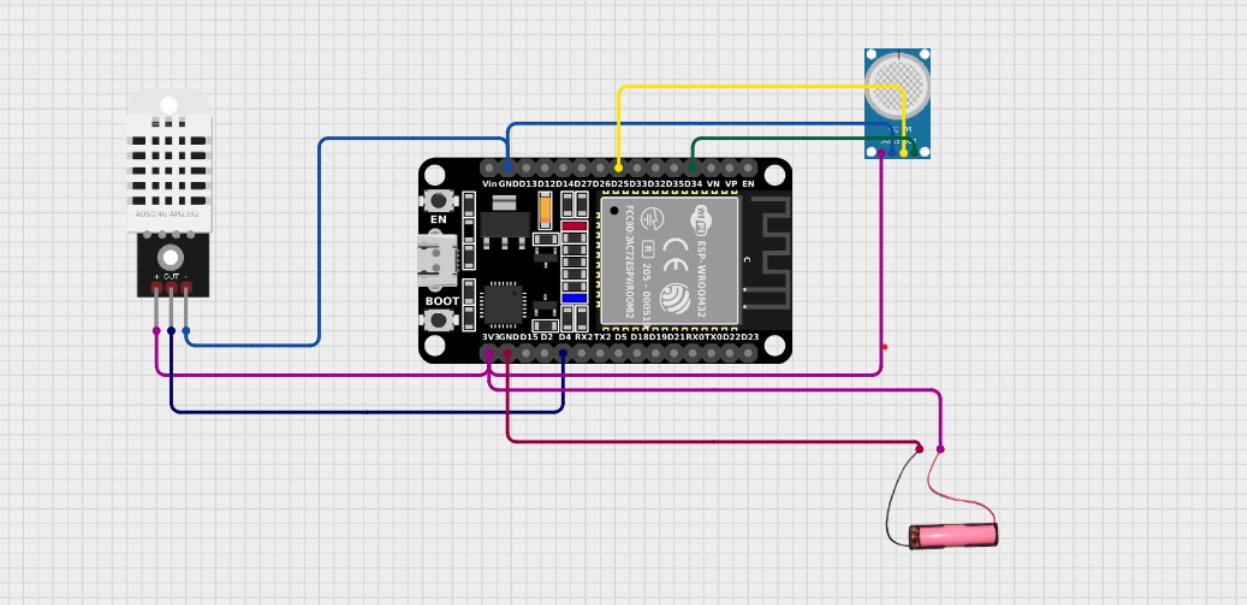


Figure: block diagram

Picture of implementation hardware and software:

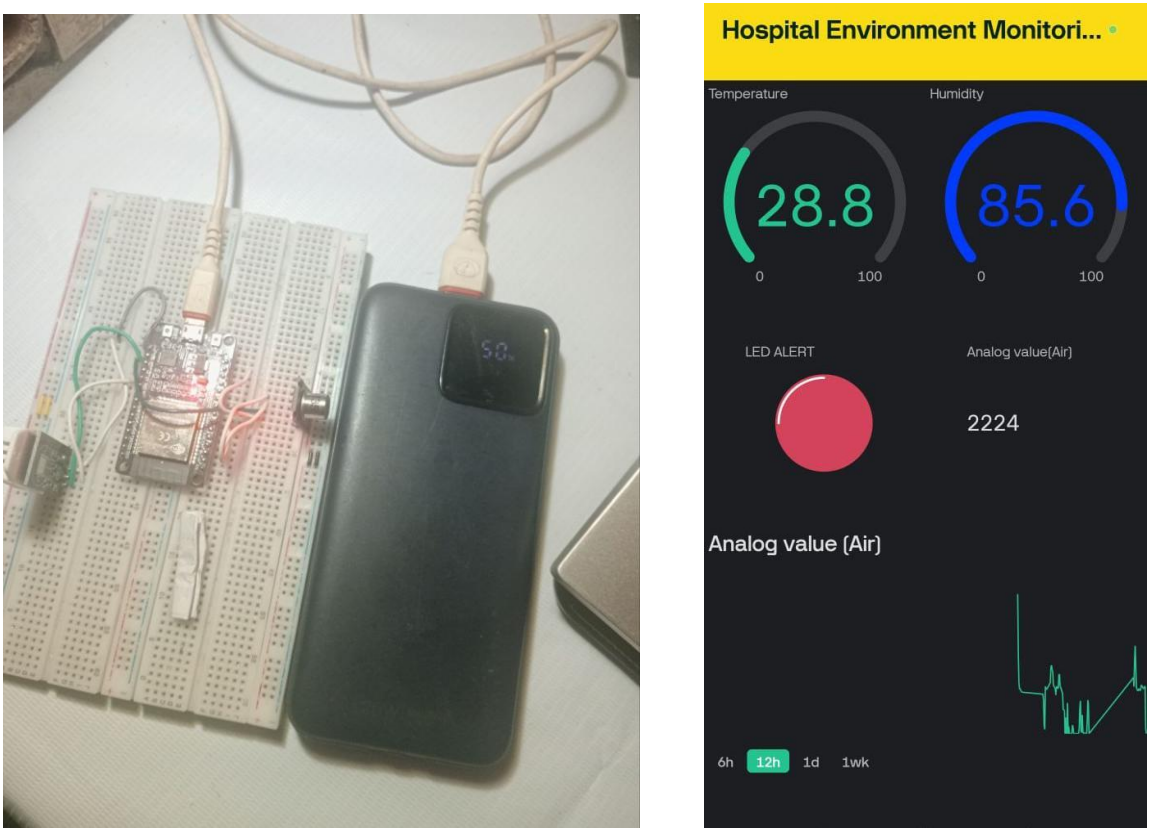


Figure: Implementation Sample Picture (Hardware & Software)

Future work:

- Add more advanced sensors and biomedical integration to improve hospital environmental and patient monitoring, including detection of harmful gases, anesthetic leaks, and wearable-based health tracking
- Use AI and machine learning for predictive analysis, enabling early detection of unsafe conditions, energy optimization, and intelligent automation of control devices.
- Expand the system to multiple hospital departments with cloud-based monitoring, centralized dashboards, and mobile access for doctors and healthcare staff.
- Enhance data security with encrypted communication protocols, ensuring patient and hospital privacy, while aligning with ISO and WHO medical standards.
- Integrate renewable energy sources such as solar backup and AI-driven smart alerts to ensure efficient, sustainable, and uninterrupted operation in critical hospital areas.

Conclusion:

This project successfully demonstrates an IoT-based hospital environmental monitoring system that ensures patient safety and staff comfort by tracking temperature, humidity, and air quality. With real-time alerts and centralized monitoring, it improves healthcare efficiency while offering scope for future integration of AI, automation, and secure large-scale deployment.

Reference:

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